Technical Paper

Paul Wayne

5/6/2025

Big Ski MG

Dear Paul,

As you may have heard, our current models of skis have been prone to breaking and deteriorating quicker than we would like due to the properties of the composite materials we are using. Additionally, binding mount tear out has been a common complaint from customers. The performance and longevity of composite skis are critically affected by mechanical deterioration over time.

Our current ski models, which employ fiber reinforced polymers (FRPs) like carbon fiber and fiberglass, are valued for their exceptional strength-to-weight ratios, flexibility, and tunable stiffness. However, these materials face progressive damage due to repeated bending, torsional stress, impact forces, and environmental exposure to moisture and temperature fluctuations which as you know is what our skis endure regularly in their product lifetime.

Delamination, microcracking, and fiber-matrix debonding arise as primary damage mechanisms under cyclic loading. As Salavatian, Fotouhi, and Soutis (2023) note, even before visible damage appears, internal weakening can reduce composite stiffness and strength, ultimately risking sudden failure [1].

- Stakeholders impacted include:
- Consumers, who face safety hazards and declining performance.
- Manufacturing teams, responsible for warranty costs and redesigns.
- Retailers, who have experienced diminished customer trust and complex inventory management. • Environmental bodies, since shorter ski lifespans increase material waste and manufacturing emissions.

Addressing these degradation issues will improve safety, extend product lifespan, reduce warranty claim numbers, and elevate our brand's reputation for durable, high-performance products.

To improve the mechanical durability of our skis, I propose the following material interventions. Firstly, the incorporation of short carbon or Kevlar fibers. By adding short carbon or Kevlar fibers within the composite laminate can substantially increase fracture toughness and improve resistance to delamination. This improves energy dissipation during impact events (like rock strikes or jump landings) and reduces the risk of interfacial crack propagation.

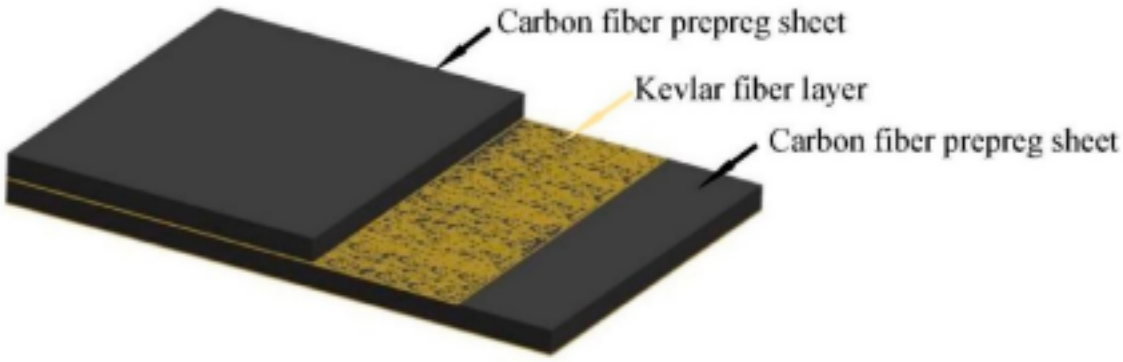


Fig 1. Kevlar-fiber integration within the structure of a composite laminate to increase fracture toughness. The structure has a carbon fiber prepreg sheet at the top and bottom and a Kevlar-fiber layer in the middle [6].

Another option is the use of nanoparticle modified epoxy systems. Nanoparticles (such as carbon nanofibers or silica particles) embedded within the epoxy matrix enhance the interlaminar fracture toughness and low-velocity impact damage tolerance. These modifications increase both fatigue strength and toughness without significantly compromising stiffness.

My final idea is moisture resistant epoxy systems. Switching to an epoxy resin with enhanced moisture resistance will improve the composite's mechanical stability under fluctuating environmental conditions. This reduces the long-term weakening effects from water ingress, which degrade fiber-matrix bonding.

Additionally, as a design change to combat the binding tear out issue, I propose that we incorporate carbon nanotubes into our composite material in the mounting region of the ski to reduce bolt tear out. This is achieved by reinforcing the composite matrix and the interface with CNTs, making it more resistant to cracking and tear-out under tensile loading [3].

There are some potential drawbacks to these design modifications. While these materials offer clear mechanical advantages, they introduce challenges such as increased material and processing costs (especially with nanofiller systems). These costs would likely be borne by our consumers and should be mitigated as much as possible to ensure our brand remains accessible to the average consumer.

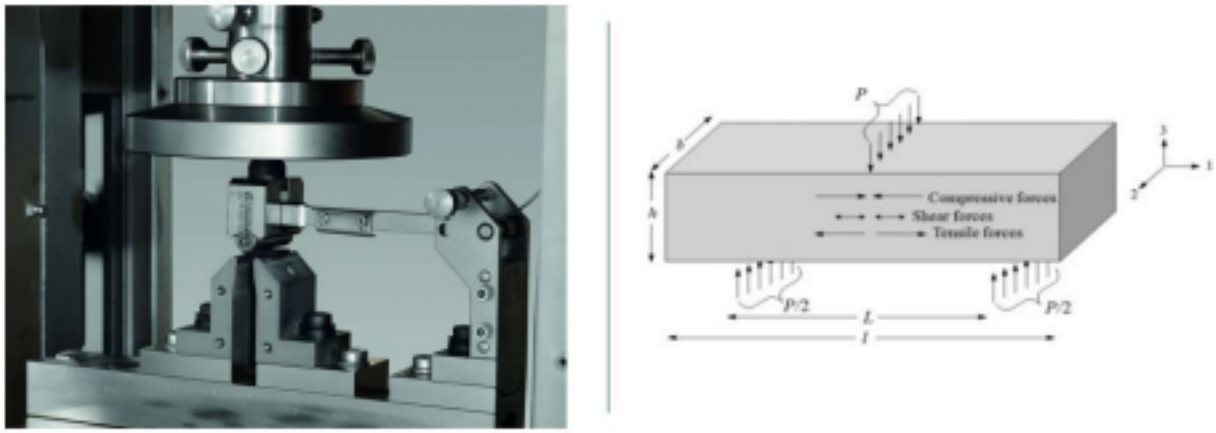
Additionally, there are potential changes in flexural behavior, requiring careful tuning to maintain desired ski's feel and responsiveness. By changing the composite materials we use, the feel of our skis will be altered, and field testing will be necessary to ensure that we don't compromise how well our skis perform.

Also adjusted manufacturing processes come with issues, as nanocomposites may require modified curing cycles or dispersion techniques. This also comes with the additional setback of needing to train the manufacturing team in these techniques or hiring new members with previous experience in these manufacturing methods.

There are a few tests that I believe would be valuable in evaluating the mechanical improvements of the proposed materials. I suggest the following testing methods:

Interlaminar shear strength (ILSS) using the ASTM D2344 standard. This test assesses the

delamination resistance at ply interfaces and obtains the shear strength between laminate planes of composite using a short beam shear test [4].



(a) (b)

Fig 2. Experimental testing for ILSS adhering to ASTM D2344 standards. (a) ILSS testing machine [4]. (b) Simplification of testing method [5].

Another test we should perform is fatigue testing (Cyclic Flexural Load) using the ASTM D7774 standard. This test is used to determine fatigue life under repeated bending loads.

The moisture absorption & retention test would also be valuable to employ. This uses the ASTM D570 standard and evaluates moisture uptake and its effect on mechanical properties.

Finally, low velocity impact resistance testing should be done with the ASTM D7136 standard to simulate and measure damage tolerance to typical rock or ice impacts.

These tests will provide quantitative data on fracture toughness, fatigue resistance, moisture durability, and impact toughness, which will allow us to compare current and modified designs so long as we test before and after design modifications. Based on these results, we can adjust laminate thicknesses, fiber orientations, and resin formulas to balance stiffness, toughness, and weight according to how we want our skis to perform.

Addressing the durability and performance issues in our current ski models is essential for maintaining customer satisfaction, brand reputation, and operational efficiency. The proposed enhancements including the integration of short Kevlar or carbon fibers, nanoparticle-modified epoxy systems, and moisture-resistant resins offer promising solutions to mitigate mechanical degradation and binding mount tear-out. While these modifications introduce new challenges in cost and manufacturing complexity, the potential improvements in safety, lifespan, and reliability justify further exploration.

By conducting rigorous mechanical testing using established ASTM standards, we can quantify the benefits and make informed adjustments to optimize ski performance. Implementing these innovations will position Big Ski MG at the forefront of durable, high-performance ski design. Additionally, any input from your expertise on these potential solutions and testing methods would be greatly appreciated. Thank you for your support and insight.

Sources

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